

Session M1 — From Exponents to Factoring Quadratics (Grade 9 Foundation)

Subject: Mathematics **Duration:** ~75 minutes (flexible 60–90 min) **Student:** Grade 9, concurrently taking an accelerated Grade 11 Math course **Topic:** Quadratic equations — building the Grade-9 foundation (exponents → polynomial factorization → factoring & solving quadratics)

Source material used tonight: R.D. Sharma *Mathematics – IX*, Chapter 2 “Exponents of Real Numbers” and Chapter 6 “Factorization of Polynomials” (the two excerpts provided).

Grade-10 / Grade-11 material: Not yet provided. This session deliberately stops at the *factorization method* for solving quadratic equations. Completing the square, the quadratic formula (and its derivation), the discriminant/nature of roots, and any Grade-11-level extensions (complex roots, sum & product of roots, forming equations from roots) belong in the **next** session once the Grade-10 and Grade-11 PDFs are available — see “Notes for Next Session” at the end.

Why this scope, tonight: The school’s accelerated course has reached quadratic equations. Before he can follow that at grade-11 depth, he needs to be completely fluent in two grade-9 building blocks: (1) the laws of exponents (so algebraic manipulation doesn’t slow him down), and (2) factoring a quadratic trinomial by splitting the middle term (the method almost every “solve the quadratic” problem starts with). This session builds both, then fuses them into “solve a quadratic equation by factorization.”

Lesson Plan

Segment	Time	Focus
Warm-up	10 min	Recall laws of exponents; hook problem that hides a quadratic equation inside an exponential equation
Concept Notes — Part A	10 min	Polynomials refresher: definition, degree, zero/root of a polynomial
Concept Notes — Part B	10 min	Remainder Theorem and Factor Theorem (statement + proof of each)
Concept Notes — Part C	15 min	Splitting the middle term — full derivation of <i>why</i> the method works
Concept Notes — Part D	5 min	Zero Product Property; definition of a quadratic equation; solving by factorization
Worked Examples	10 min	3 fully solved examples, increasing in difficulty
Practice Problems	15 min (in-class) / homework	13 problems across 4 tiers
Wrap-up	5 min	Answer key walkthrough, preview of next session

Learning objectives — by the end of this session the student will be able to:

1. State and correctly apply all the laws of exponents.
2. State the Remainder Theorem and Factor Theorem, and explain *why* each is true (not just recite it).

- Factor any quadratic trinomial $ax^2 + bx + c$ by splitting the middle term, and explain why the method works.
- Solve a quadratic equation $ax^2 + bx + c = 0$ by factorization, using the Zero Product Property.
- Recognize and solve an exponential equation that is secretly a quadratic equation in disguise (via substitution).

Warm-up (10 min)

Quick recall: Laws of Exponents

Before we start, run through these out loud — for each one, be ready to say *why* it's true, not just what it says (we derive all of them in the notes below):

$$a^m \times a^n = a^{m+n} \quad a^m \div a^n = a^{m-n} \quad (a^m)^n = a^{mn}$$

$$(ab)^m = a^m b^m \quad \left(\frac{a}{b}\right)^m = \frac{a^m}{b^m} \quad a^0 = 1 \quad a^{-n} = \frac{1}{a^n}$$

Quick fire (answer out loud, 2 min):

- $2^3 \times 2^4 = ? \rightarrow 2^7 = 128$
- $5^0 = ? \rightarrow 1$
- $3^{-2} = ? \rightarrow \frac{1}{9}$

Hook problem

Solve for x : $2^{2x+1} = 17 \cdot 2^x - 8$

This doesn't look like anything we've solved before — there's an x in the exponent *and* the equation isn't linear. Watch what happens if we let $y = 2^x$:

$$2 \cdot (2^x)^2 = 17 \cdot 2^x - 8 \implies 2y^2 - 17y + 8 = 0$$

That's a **quadratic equation** in y . We don't yet have a general, reliable way to solve *any* quadratic equation — that's exactly what tonight builds. We'll come back and finish this exact problem in Worked Example 3.

Concept Notes

Part A — Polynomials Refresher (10 min)

Definition (Polynomial). An algebraic expression of the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

where n is a non-negative integer and the coefficients $a_n, a_{n-1}, \dots, a_0$ are real numbers with $a_n \neq 0$, is called a **polynomial in x of degree n** .

Degree names you'll see constantly: degree 1 $\rightarrow$ *linear*, degree 2 $\rightarrow$ *quadratic*, degree 3 $\rightarrow$ *cubic*, degree 4 $\rightarrow$ *biquadratic* (*quartic*).

Definition (Quadratic polynomial). A polynomial of degree 2: $ax^2 + bx + c$, where a, b, c are real numbers and $a \neq 0$. (Why $a \neq 0$? If $a = 0$ the x^2 term vanishes and it's just a linear expression $bx + c$ — not

degree 2 anymore.)

Definition (Zero / Root of a polynomial). A real number k is called a **zero** (or **root**) of a polynomial $f(x)$ if $f(k) = 0$.

This is the idea we'll use constantly tonight: "finding where a polynomial equals zero" and "finding its factors" turn out to be the same question — that's exactly what the next two theorems formalize.

Part B — Remainder Theorem & Factor Theorem (10 min)

Remainder Theorem. If a polynomial $f(x)$ is divided by a linear polynomial $(x - a)$, the remainder is $f(a)$.
Mathematically: $f(x) = (x - a) \cdot q(x) + f(a)$, for some polynomial $q(x)$ (the quotient).

Derivation (why this is true): When you divide any polynomial $f(x)$ by a linear divisor $(x - a)$, the division algorithm guarantees

$$f(x) = (x - a) \cdot q(x) + r$$

where r is the remainder. Since the divisor $(x - a)$ has degree 1, the remainder must have degree *less than* 1 — i.e., r is just a constant number (not an expression in x).

Now substitute $x = a$ into both sides:

$$f(a) = (a - a) \cdot q(a) + r = 0 \cdot q(a) + r = r$$

So $r = f(a)$. That's the whole proof — the remainder is literally just "plug a into f ." ■

Factor Theorem. $(x - a)$ is a factor of a polynomial $f(x)$ **if and only if** $f(a) = 0$.

Derivation (this follows directly from the Remainder Theorem):

- ($\Leftarrow$) Suppose $f(a) = 0$. From the Remainder Theorem, $f(x) = (x - a)q(x) + f(a) = (x - a)q(x) + 0 = (x - a)q(x)$. Since $f(x)$ divides evenly by $(x - a)$ with nothing left over, $(x - a)$ is a factor of $f(x)$.
- ($\Rightarrow$) Suppose $(x - a)$ is a factor of $f(x)$. Then $f(x) = (x - a)q(x)$ for some $q(x)$. Substituting $x = a$: $f(a) = (a - a)q(a) = 0$.

Both directions hold, so it's an "if and only if." ■

Quick example: Is $(x - 2)$ a factor of $f(x) = x^3 - 6x^2 + 11x - 6$? $f(2) = 8 - 24 + 22 - 6 = 0 \rightarrow$ yes, by the Factor Theorem, $(x - 2)$ is a factor.

Part C — Splitting the Middle Term: the full derivation (15 min)

This is the core technique tonight, and the standing rule in this course is: never memorize a method without knowing why it works. So here's where "find two numbers whose sum is b and product is ac " actually comes from.

Goal: factor a quadratic trinomial $ax^2 + bx + c$ (with a, b, c real, $a \neq 0$) into two linear factors $(px + q)(rx + s)$.

Derivation:

Step 1 — Expand the general product of two linear factors:

$$(px + q)(rx + s) = prx^2 + (ps + qr)x + qs$$

Step 2 — Match coefficients with $ax^2 + bx + c$:

$$pr = a \quad qs = c \quad ps + qr = b$$

Step 3 — Here's the key trick. Let $m = ps$ and $n = qr$, so that $m + n = b$ (matches the middle coefficient, by the third equation above). Now look at the *product* $m \cdot n$:

$$m \cdot n = (ps)(qr) = (pr)(qs) = a \cdot c$$

So m and n are two numbers whose **sum is b** and whose **product is ac** . That's exactly the rule — and now you can see it isn't an arbitrary trick, it falls straight out of matching coefficients.

Step 4 — Once such m, n are found, rewrite the middle term bx as $mx + nx$:

$$ax^2 + bx + c = ax^2 + mx + nx + c$$

Step 5 — Factor by grouping: pull the common factor out of the first pair of terms, and out of the second pair, and you'll always be left with the *same* binomial factor in both — that's what lets you finish the factoring.

Worked derivation example: Factor $6x^2 + 7x - 3$.

- $a = 6, b = 7, c = -3 \Rightarrow ac = -18$.
- Need two numbers with sum 7 and product -18 : try 9 and -2 ($9 \times (-2) = -18, 9 + (-2) = 7$). ✓
- Split: $6x^2 + 9x - 2x - 3$
- Group: $3x(2x + 3) - 1(2x + 3)$
- Common factor: $(2x + 3)(3x - 1)$

Check by expanding: $(2x + 3)(3x - 1) = 6x^2 - 2x + 9x - 3 = 6x^2 + 7x - 3$ ✓.

This is exactly the technique the textbook uses to finish factoring higher-degree polynomials too — e.g. after using the Factor Theorem to strip a linear factor off a cubic like $2x^3 - 3x^2 - 17x + 30$, you're left with a quadratic quotient ($2x^2 + x - 15$), and splitting the middle term finishes the job. We'll solve that exact quotient as an equation in Worked Example 2.

Part D — Zero Product Property & Solving Quadratic Equations (5 min)

Zero Product Property. For real numbers p and q : if $p \cdot q = 0$, then $p = 0$ or $q = 0$ (or both).

Mathematically: $pq = 0 \iff p = 0 \vee q = 0$

Reasoning: Suppose $p \neq 0$. Then we're allowed to divide both sides of $pq = 0$ by p : $q = \frac{0}{p} = 0$. So if p isn't zero, q is forced to be zero — meaning at least one of the two factors must be zero. This is the property that lets us turn a *factored* equation into two simple linear equations.

Definition (Quadratic equation). An equation of the form $ax^2 + bx + c = 0$, where a, b, c are real numbers and $a \neq 0$.

Solving by factorization — putting it all together:

1. Factor $ax^2 + bx + c$ into $(px + q)(rx + s)$ by splitting the middle term (Part C).
2. Set the factored equation to zero: $(px + q)(rx + s) = 0$.
3. Apply the Zero Product Property: $px + q = 0$ **or** $rx + s = 0$.
4. Solve each linear equation: $x = -\frac{q}{p}$ or $x = -\frac{s}{r}$.

Note for next session: factorization only works cleanly when the quadratic splits into nice

integer/rational factors. When it doesn't, we need the completing-the-square method and the quadratic formula — that's Grade-10 territory, coming once that material is available.

Worked Examples (10 min)

Worked Example 1 — Factor a quadratic trinomial

Factor: $2x^2 + 5x - 3$

$a = 2, b = 5, c = -3 \Rightarrow ac = -6$. Need two numbers with sum 5, product -6 : 6 and -1 ($6 \times (-1) = -6, 6 + (-1) = 5$). ✓

$$2x^2 + 6x - x - 3 = 2x(x + 3) - 1(x + 3) = (x + 3)(2x - 1)$$

$$\text{Check: } (x + 3)(2x - 1) = 2x^2 - x + 6x - 3 = 2x^2 + 5x - 3 \quad \checkmark$$

Worked Example 2 — Solve a quadratic equation by factorization

Solve: $2x^2 + x - 15 = 0$

(This is the exact quadratic quotient that appears inside the textbook's Example 1 of §6.5, when factoring $2x^3 - 3x^2 - 17x + 30$ — now we solve it as an equation in its own right.)

$a = 2, b = 1, c = -15 \Rightarrow ac = -30$. Need sum 1, product -30 : 6 and -5 ($6 \times (-5) = -30, 6 + (-5) = 1$). ✓

$$2x^2 + 6x - 5x - 15 = 2x(x + 3) - 5(x + 3) = (x + 3)(2x - 5)$$

$$\text{Set to zero: } (x + 3)(2x - 5) = 0 \Rightarrow x + 3 = 0 \text{ or } 2x - 5 = 0 \Rightarrow x = -3 \text{ or } x = \frac{5}{2}$$

$$\text{Check } x = -3: 2(9) + (-3) - 15 = 18 - 3 - 15 = 0 \quad \checkmark \quad \text{Check } x = \frac{5}{2}: 2\left(\frac{25}{4}\right) + \frac{5}{2} - 15 = \frac{25}{2} + \frac{5}{2} - 15 = \frac{30}{2} - 15 = 15 - 15 = 0 \quad \checkmark$$

Worked Example 3 — Finishing the warm-up hook

Solve: $2^{2x+1} = 17 \cdot 2^x - 8$

Rewrite $2^{2x+1} = 2 \cdot (2^x)^2$. Let $y = 2^x$:

$$2y^2 = 17y - 8 \implies 2y^2 - 17y + 8 = 0$$

$a = 2, b = -17, c = 8 \Rightarrow ac = 16$. Need sum -17 , product 16: -16 and -1 ($-16 \times -1 = 16, -16 + (-1) = -17$). ✓

$$2y^2 - 16y - y + 8 = 2y(y - 8) - 1(y - 8) = (y - 8)(2y - 1)$$

$$(y - 8)(2y - 1) = 0 \Rightarrow y = 8 \text{ or } y = \frac{1}{2}$$

Back-substitute $y = 2^x$:

- $2^x = 8 = 2^3 \Rightarrow x = 3$
- $2^x = \frac{1}{2} = 2^{-1} \Rightarrow x = -1$

Check $x = 3$: LHS = $2^7 = 128$. RHS = $17(8) - 8 = 136 - 8 = 128$ ✓ Check $x = -1$: LHS = $2^{-1} = 0.5$. RHS = $17(0.5) - 8 = 8.5 - 8 = 0.5$ ✓

$$x = 3 \text{ or } x = -1$$

Practice Problems

Tier 1 — Laws of Exponents (warm-up level)

1. Simplify: $\frac{2^5 \times 2^{-2}}{2^{-3}}$
2. Solve for x : $3^{x+2} = 81$
3. Simplify: $(a^3b^{-2})^2 \times (a^{-1}b)^3$

Tier 2 — Factor by splitting the middle term

4. Factor: $4x^2 - 4x - 15$
5. Factor: $3x^2 - 5x - 12$
6. Factor: $5x^2 + 13x - 6$

Tier 3 — Solve quadratic equations by factorization

7. Solve: $x^2 - 7x + 12 = 0$
8. Solve: $2x^2 - x - 6 = 0$
9. Solve: $3x^2 + 14x - 5 = 0$
10. Solve: $6x^2 - x - 15 = 0$

Tier 4 — Challenge: exponential equations hiding a quadratic

11. Solve for x : $2^{2x} - 9(2^x) + 8 = 0$
12. Solve for x : $3^{2x+1} - 10(3^x) + 3 = 0$

Tier 5 — Apply the Factor Theorem (quick check, ties Part B back in)

13. Without dividing, determine whether $(x - 1)$ is a factor of $f(x) = x^3 + 2x^2 - x - 2$. If it is, use the Factor Theorem's logic plus splitting the middle term (after dividing out the known factor) to fully factor $f(x)$.

Answer Key — Full Step-by-Step Solutions

1. $\frac{2^5 \times 2^{-2}}{2^{-3}} = 2^{5+(-2)-(-3)} = 2^{5-2+3} = 2^6 = 64$

2. $3^{x+2} = 81 = 3^4 \Rightarrow x + 2 = 4 \Rightarrow x = 2$

3. $(a^3b^{-2})^2 \times (a^{-1}b)^3 = a^6b^{-4} \times a^{-3}b^3 = a^{6-3}b^{-4+3} = a^3b^{-1} = \frac{a^3}{b}$

4. Factor $4x^2 - 4x - 15$: $a = 4, c = -15 \Rightarrow ac = -60$. Need sum -4 , product -60 : $6, -10$ ($6 \times -10 = -60, 6 - 10 = -4$). $\checkmark$ $4x^2 + 6x - 10x - 15 = 2x(2x + 3) - 5(2x + 3) = (2x + 3)(2x - 5)$ Check: $4x^2 - 10x + 6x - 15 = 4x^2 - 4x - 15 \checkmark$

5. Factor $3x^2 - 5x - 12$: $ac = -36$. Need sum -5 , product -36 : $4, -9$ ($4 \times -9 = -36, 4 - 9 = -5$). $\checkmark$ $3x^2 + 4x - 9x - 12 = x(3x + 4) - 3(3x + 4) = (3x + 4)(x - 3)$ Check: $3x^2 - 9x + 4x - 12 = 3x^2 - 5x - 12 \checkmark$

6. Factor $5x^2 + 13x - 6$: $ac = -30$. Need sum 13 , product -30 : $15, -2$ ($15 \times -2 = -30, 15 - 2 = 13$). $\checkmark$ $5x^2 + 15x - 2x - 6 = 5x(x + 3) - 2(x + 3) = (x + 3)(5x - 2)$ Check: $5x^2 - 2x + 15x - 6 = 5x^2 + 13x - 6 \checkmark$

7. Solve $x^2 - 7x + 12 = 0$: need sum -7 , product 12 : $-3, -4$. $(x - 3)(x - 4) = 0 \Rightarrow x = 3$ or $x = 4$
Check: $x = 3$: $9 - 21 + 12 = 0 \checkmark$. $x = 4$: $16 - 28 + 12 = 0 \checkmark$

8. Solve $2x^2 - x - 6 = 0$: $ac = -12$. Need sum -1 , product -12 : $3, -4$. $2x^2 + 3x - 4x - 6 = x(2x + 3) - 2(2x + 3) = (2x + 3)(x - 2)$ Check: $2x^2 - 4x + 3x - 6 = 2x^2 - x - 6 \checkmark$

$$3) - 2(2x + 3) = (2x + 3)(x - 2) \quad (2x + 3)(x - 2) = 0 \Rightarrow x = -\frac{3}{2} \text{ or } x = 2 \text{ Check: } x = 2: 8 - 2 - 6 = 0 \checkmark. \quad x = -\frac{3}{2}: 2\left(\frac{9}{4}\right) + \frac{3}{2} - 6 = 4.5 + 1.5 - 6 = 0 \checkmark$$

9. Solve $3x^2 + 14x - 5 = 0$: $ac = -15$. Need sum 14, product -15 : 15, -1 . $3x^2 + 15x - x - 5 = 3x(x + 5) - 1(x + 5) = (x + 5)(3x - 1) \quad (x + 5)(3x - 1) = 0 \Rightarrow x = -5 \text{ or } x = \frac{1}{3}$ Check: $x = -5$: $75 - 70 - 5 = 0 \checkmark$. $x = \frac{1}{3}$: $3\left(\frac{1}{9}\right) + \frac{14}{3} - 5 = \frac{1}{3} + \frac{14}{3} - 5 = \frac{15}{3} - 5 = 0 \checkmark$

10. Solve $6x^2 - x - 15 = 0$: $ac = -90$. Need sum -1 , product -90 : 9, -10 . $6x^2 + 9x - 10x - 15 = 3x(2x + 3) - 5(2x + 3) = (2x + 3)(3x - 5) \quad (2x + 3)(3x - 5) = 0 \Rightarrow x = -\frac{3}{2} \text{ or } x = \frac{5}{3}$ Check: $x = \frac{5}{3}$: $6\left(\frac{25}{9}\right) - \frac{5}{3} - 15 = \frac{50}{3} - \frac{5}{3} - 15 = \frac{45}{3} - 15 = 15 - 15 = 0 \checkmark$. $x = -\frac{3}{2}$: $6\left(\frac{9}{4}\right) + \frac{3}{2} - 15 = 13.5 + 1.5 - 15 = 0 \checkmark$

11. Solve $2^{2x} - 9(2^x) + 8 = 0$: let $y = 2^x$: $y^2 - 9y + 8 = 0 \Rightarrow (y - 1)(y - 8) = 0 \Rightarrow y = 1 \text{ or } y = 8$
 $2^x = 1 = 2^0 \Rightarrow x = 0$ $2^x = 8 = 2^3 \Rightarrow x = 3$ Check $x = 0$: $2^0 - 9(2^0) + 8 = 1 - 9 + 8 = 0 \checkmark$. Check $x = 3$: $2^6 - 9(2^3) + 8 = 64 - 72 + 8 = 0 \checkmark$

12. Solve $3^{2x+1} - 10(3^x) + 3 = 0$: rewrite $3^{2x+1} = 3 \cdot (3^x)^2$. Let $y = 3^x$: $3y^2 - 10y + 3 = 0$. $ac = 9$. Need sum -10 , product 9: $-9, -1$. $3y^2 - 9y - y + 3 = 3y(y - 3) - 1(y - 3) = (y - 3)(3y - 1) = 0 \Rightarrow y = 3 \text{ or } y = \frac{1}{3}$ $3^x = 3 = 3^1 \Rightarrow x = 1$ $3^x = \frac{1}{3} = 3^{-1} \Rightarrow x = -1$ Check $x = 1$: $3^3 - 10(3) + 3 = 27 - 30 + 3 = 0 \checkmark$. Check $x = -1$: $3^{-1} - 10(3^{-1}) + 3 = \frac{1}{3} - \frac{10}{3} + 3 = -\frac{9}{3} + 3 = -3 + 3 = 0 \checkmark$

13. Is $(x - 1)$ a factor of $f(x) = x^3 + 2x^2 - x - 2$? By the Factor Theorem, check $f(1)$: $f(1) = 1 + 2 - 1 - 2 = 0$. Since $f(1) = 0$, yes $-(x - 1)$ is a factor.

Divide $f(x)$ by $(x - 1)$ (long division) to get the quadratic quotient: $x^3 + 2x^2 - x - 2 \div (x - 1) = x^2 + 3x + 2$

So $f(x) = (x - 1)(x^2 + 3x + 2)$.

Now factor $x^2 + 3x + 2$ by splitting the middle term: need sum 3, product 2: 1, 2. $x^2 + x + 2x + 2 = x(x + 1) + 2(x + 1) = (x + 1)(x + 2)$

Therefore: $f(x) = (x - 1)(x + 1)(x + 2)$

Check by expanding $(x - 1)(x + 1)(x + 2)$: $(x - 1)(x + 1) = x^2 - 1$, then $(x^2 - 1)(x + 2) = x^3 + 2x^2 - x - 2 \checkmark$

Notes for Next Session

Tonight covered the complete grade-9 foundation: laws of exponents, the Remainder/Factor Theorems, and factoring & solving quadratic equations by splitting the middle term — matched to what his accelerated course is currently doing.

Still pending (once the grade-10 and grade-11 R.D. Sharma PDFs are available):

- **Grade 10 — Quadratic Equations chapter:** the completing-the-square method (with full derivation), the quadratic formula (derived from completing the square, not just stated), and the discriminant $b^2 - 4ac$ for determining the nature of roots (real & distinct, real & equal, no real roots).
- **Grade 11 — accelerated depth:** likely complex/imaginary roots when the discriminant is negative, sum and product of roots ($\alpha + \beta = -b/a$, $\alpha\beta = c/a$) and forming a quadratic from given roots, and possibly quadratic inequalities.

Once those two chapters are shared, the plan is to build the continuation session(s) picking up exactly where tonight left off — same 9→10→11 interleaved structure used for the Physics track.